

Deepak Mallampati

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Kent, OH, USA - open to international relocation

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) building enterprise AI agent deployments - Healthcare RAG with ~35% retrieval precision, agentic LLM workflows with >30% hallucination reduction, schema-validated multi-agent claims pipelines, and FastAPI / Docker production services with ~30% post-deployment defect reduction. Solo-shipped Manga Lens to the Chrome Web Store integrating four AI vision providers. Comfortable owning customer-facing AI agent deployments in regulated domains.

SKILLS

Forward Deployed Engineering • Enterprise AI Agent Deployment • Conversational AI • LLM Applications & RAG • Python / JavaScript / TypeScript • Data Modeling & Integration • Customer-Facing Delivery • Production Reliability

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++, PHP

LLM & GenAI: LLM application development, RAG, agentic workflows, structured outputs, guardrails, grounding, evaluation pipelines

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex

Infra & DevOps: Docker, Jenkins CI/CD, Grafana, RESTful APIs, structured logging

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - USA (Healthcare Technology, AI Consulting)

- Built **RAG systems** for clinical knowledge retrieval; ~35% retrieval precision gain, >90% grounded-response alignment.
- Developed **agentic LLM workflows** for multi-step healthcare queries; agent stability +~25%, hallucination reduction >30%.
- Packaged ML/LLM inference as **FastAPI/Flask + Docker**; post-deployment defects ~30%.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Hyderabad, India

- Built **CI/CD pipelines with Jenkins**; deployment errors ->30%, release cycle -35-40%.

PROJECTS

Manga Lens - Chrome Extension (shipped, Chrome Web Store)

Solo-shipped Chrome MV3 extension integrating four AI vision providers with per-provider payload handling.

Agentic Healthcare Claims Processing System

Multi-agent pipeline with schema-validated JSON contracts; RAG-grounded validation; audit-ready reasoning traces.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025